Circles and Parts of Circles

Name _____ Date _____ Calculator allowed Version 1

Scan to practise this topic online

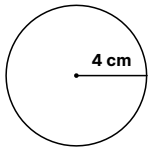
KEY WORDS radius *(pūtoro)* · diameter *(whitianga)* · circumference *(paenga porowhita)* · pi

A Circumference and area ●○○

Remember: Circumference $C = 2\pi r = \pi d$. Area $A = \pi r^2$: square the radius first. Use the π key and round at the end.

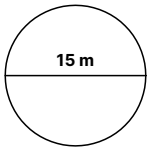
1. Find the circumference or the area, to 1 decimal place.

a)



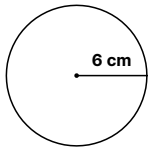
$C =$ _____

b)



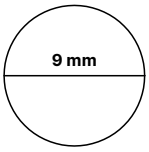
$C =$ _____

c)



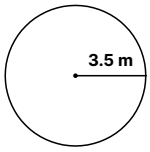
$A =$ _____

d)



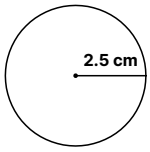
$A =$ _____

e)



$C =$ _____

f)



$A =$ _____

B Working backwards ●●○

Remember: From a circumference: $r = C \div 2\pi$. From an area: $r = \sqrt{A \div \pi}$. The diameter is twice the radius.

2. Find the length asked for, to 1 decimal place.

a) The circumference is 40 cm. Find the radius.

b) The circumference is 95 m. Find the diameter.

c) The area is 50 cm². Find the radius.

d) The area is 120 mm². Find the diameter.

e) The circumference is 28.3 m. Find the diameter.

f) The area is 254.5 cm². Find the radius.

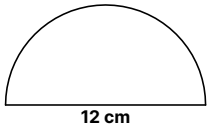
C Semicircles and quarter circles ●●○

Remember: Halve or quarter the circle's area. For the perimeter, remember to add the straight edges to the curved part.

3. A semicircle with a diameter of 12 cm.

a) Find its area.

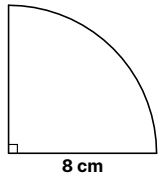
b) Find its perimeter.



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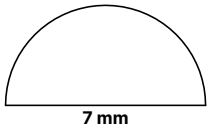
4. A quarter circle with a radius of 8 cm.

a) Find its area. **b)** Find its perimeter.



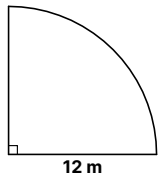
5. A semicircle with a diameter of 7 mm.

a) Find its area. **b)** Find its perimeter.



6. A quarter circle with a radius of 12 m.

a) Find its area. **b)** Find its perimeter.

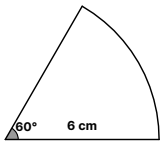


D Sectors ●●●

Remember: A sector is a fraction of a circle: the fraction is $\frac{\text{angle}}{360}$. Arc length = that fraction of the circumference; area = that fraction of the circle's area.

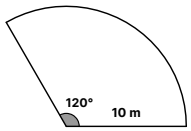
7. A sector of a circle with radius 6 cm and angle 60°.

a) Find the length of the arc. **b)** Find the area of the sector.



8. A sector of a circle with radius 10 m and angle 120°.

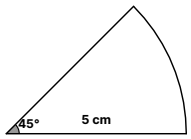
a) Find the length of the arc. **b)** Find the area of the sector.



9. A sector of a circle with radius 5 cm and angle 45°.

a) Find the length of the arc. **b)** Find the area of the sector.

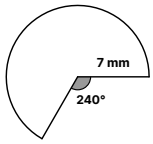
c) Find the perimeter of the sector.



10. A sector of a circle with radius 7 mm and angle 240°.

a) Find the length of the arc. **b)** Find the area of the sector.

c) Find the perimeter of the sector.

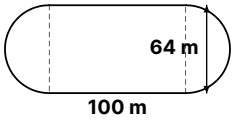


E Problems ●●●

11. A running track has two straight sides of 100 m and two semicircular ends of diameter 64 m.

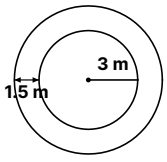
a) Find the distance around the track.

b) Find the area inside the track.



12. A circular pond of radius 3 m has a path 1.5 m wide all the way round it.

a) Find the area of the path.



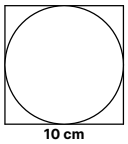
13. A bicycle wheel has a diameter of 66 cm. How many complete turns does it make on a ride of 1 km?

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14. A pizza of diameter 30 cm is cut into 8 equal slices. Find the area of one slice.

Extension

15. A square of side 10 cm has the largest possible circle drawn inside it. Find the shaded area outside the circle.



16. A sector of a circle with radius 8 cm has an area of 20 cm². Find the angle of the sector, to the nearest degree.

17. The radius of a circle is doubled. What happens to its circumference, and what happens to its area? Explain.

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Circles and Parts of Circles

Name _____ Date _____ Calculator allowed Version 2

Scan to practise this topic online

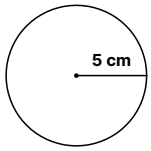
KEY WORDS radius *(pūtoro)* · diameter *(whitianga)* · circumference *(paenga porowhita)* · pi

A Circumference and area ●○○

Remember: Circumference $C = 2\pi r = \pi d$. Area $A = \pi r^2$: square the radius first. Use the π key and round at the end.

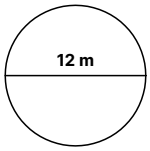
1. Find the circumference or the area, to 1 decimal place.

a)



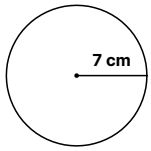
$C =$ _____

b)



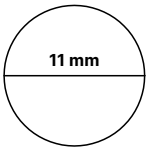
$C =$ _____

c)



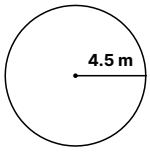
$A =$ _____

d)



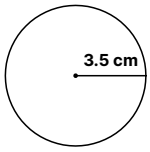
$A =$ _____

e)



$C =$ _____

f)



$A =$ _____

B Working backwards ●●○

Remember: From a circumference: $r = C \div 2\pi$. From an area: $r = \sqrt{A \div \pi}$. The diameter is twice the radius.

2. Find the length asked for, to 1 decimal place.

a) The circumference is 45 cm. Find the radius.

b) The circumference is 80 m. Find the diameter.

c) The area is 70 cm². Find the radius.

d) The area is 150 mm². Find the diameter.

e) The circumference is 31.4 m. Find the diameter.

f) The area is 201.1 cm². Find the radius.

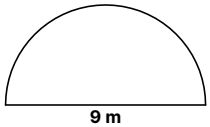
C Semicircles and quarter circles ●●○

Remember: Halve or quarter the circle's area. For the perimeter, remember to add the straight edges to the curved part.

3. A semicircle with a diameter of 9 m.

a) Find its area.

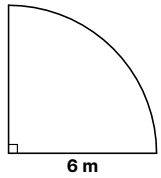
b) Find its perimeter.



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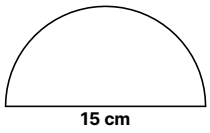
4. A quarter circle with a radius of 6 m.

a) Find its area. **b)** Find its perimeter.



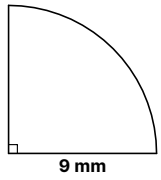
5. A semicircle with a diameter of 15 cm.

a) Find its area. **b)** Find its perimeter.



6. A quarter circle with a radius of 9 mm.

a) Find its area. **b)** Find its perimeter.

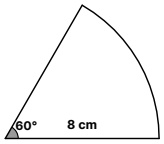


D Sectors ●●●

Remember: A sector is a fraction of a circle: the fraction is $\frac{\text{angle}}{360}$. Arc length = that fraction of the circumference; area = that fraction of the circle's area.

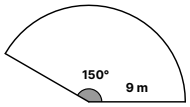
7. A sector of a circle with radius 8 cm and angle 60°.

a) Find the length of the arc. **b)** Find the area of the sector.



8. A sector of a circle with radius 9 m and angle 150°.

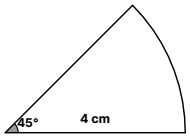
a) Find the length of the arc. **b)** Find the area of the sector.



9. A sector of a circle with radius 4 cm and angle 45°.

a) Find the length of the arc. **b)** Find the area of the sector.

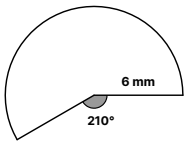
c) Find the perimeter of the sector.



10. A sector of a circle with radius 6 mm and angle 210°.

a) Find the length of the arc. **b)** Find the area of the sector.

c) Find the perimeter of the sector.

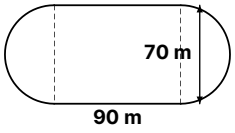


E Problems ●●●

11. A running track has two straight sides of 90 m and two semicircular ends of diameter 70 m.

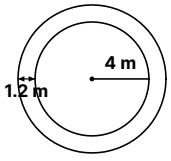
a) Find the distance around the track.

b) Find the area inside the track.



12. A circular pond of radius 4 m has a path 1.2 m wide all the way round it.

a) Find the area of the path.



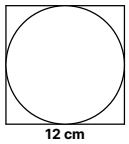
13. A bicycle wheel has a diameter of 70 cm. How many complete turns does it make on a ride of 2 km?

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14. A pizza of diameter 32 cm is cut into 6 equal slices. Find the area of one slice.

Extension

15. A square of side 12 cm has the largest possible circle drawn inside it. Find the shaded area outside the circle.



16. A sector of a circle with radius 10 cm has an area of 35 cm². Find the angle of the sector, to the nearest degree.

17. The radius of a circle is doubled. What happens to its circumference, and what happens to its area? Explain.

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